

PHYS 2801 - Assignment 2

Son Thanh Truong

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1 Question 1: The Maxwell–Boltzmann distribution

The Maxwell–Boltzmann distribution:

1.1 Part (a)

Here is the Maxwell–Boltzmann distribution function:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp \left(-\frac{mv^2}{2k_B T} \right) \quad (1)$$

Equation 1 follows from Newton’s second law. As the gravitational acceleration is g , the downward gravitation force on an object with mass m is $m g$.

To improve the readability of an equation, I usually add a half-space between the terms, which is done using back-slash followed by comma: `\,.` It looks OK also without this: $F = m\vec{a}$.

1.1.1 Complex equation

The time-independent Schrödinger equation is given by

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{r}) + V(\mathbf{r}) \psi(\mathbf{r}) = E \psi(\mathbf{r}), \quad (2)$$

where ∇^2 is the Laplacian operator, $\psi(\mathbf{r})$ is the wave function and $V(\mathbf{r})$ gives the potential energy.

1.2 Figures

Figures are included to a latex document using syntax as seen below.

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.6\textwidth]{example.png}
  \caption{An example figure. Replace the file name and caption with your own.}
  \label{fig:example}
\end{figure}
```

Comments:

- The key line is the `\includegraphics` command, which inserts the figure and allows you to specify its size.
- The image file must exist in the same folder as your `.tex` file. The recommended formats are `.png` or `.pdf`.
- To refer to the figure from the text, use its label, e.g. see `Figure~\ref{fig:example}`.

1.3 Question 2

Now onto question 2!